INCH-POUND
A-A-59457B
April 23, 2025
SUPERSEDING
A-A-59457A
July 20, 2001

COMMERCIAL ITEM DESCRIPTION

Drilling Machine, Round Column, Floor-Mounted

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted, round column, drilling machines intended for use in light – medium duty general drilling operations used in fabrication applications by Air Force (AF) Metals Technology (MT) shop installations. AF MT shops serve as the organic and local manufacturing / fabrication resource across the enterprise (150+ locations).

1.1 Intended use. The floor mounted drilling machines covered by this CID are intended for use in general-purpose drilling, tapping, boring, counterboring, spot facing, and reaming operations on large and/or heavy workpieces. The machines are intended for use in Air Force Metals Technology shops via Air Force Specialty Code (AFSC) 2A7X1.

2. CLASSIFICATION. The drilling machine shall be of the types and arm swing length as shown in table I.

3. SALIENT CHARACTERISTICS.

3.1 General requirements. The drilling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The drilling machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support a worktable, column, and drill head assembly. The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) BS.IM. The base shall have a means for leveling and anchoring the machine to the floor.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28 Wright-Patterson AFB, OH 45433 or email to engineering.standards@us.af.mil. Since contact information can change, you should verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

TABLE I. Types, max drill capacity and arm swing length of drilling machines.

Type[1]	Max drill capacity, steel, inches	Arm swing length, inches
01	1-1/4	20.8
02	1-1/4	18

[1]The machine maximum drill diameter and arm swing length shall not be less than the stated requirements of the size ordered nor greater than the next larger size shown.

[2]The arm swing length of the drilling machine is defined as the horizontal distance between the column face and the centerline of the drill head spindle at its maximum travel distance from the column.

3.2.2 Worktable. The dimensions of the worktable shall be those considered by the manufacturer as standard. The worktable shall rotate 360° about the center of the column and shall be manually adjustable vertically along the column to any position between the base and drill head assembly. A locking device shall be provided to secure the worktable after adjustment. The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) BS.IM.

3.2.3 Drill head assembly. The drill head assembly shall include a power transmission system and a quill and spindle. The drill head assembly shall rotate 360° about the center of the column. The drill head assembly shall be vertically adjustable along the column. A locking device shall be provided to secure the drill head assembly after adjustment.

3.2.3.1 Power transmission. The spindle speed range and drive motor horsepower shall meet the requirements shown in table II. The spindle speed shall be infinitely variable throughout the range.

TABLE II. Quill/spindle assembly & drive requirements.

Machine Type	Spindle speed range Low / high rpm	Motor horsepower	Spindle taper morse (MT)
01	85—540 245—2000	2	#4
02	80—1000 230—2500	2	#3

[1]The machine maximum drill diameter and radial arm length shall not be less than the stated requirements of the size ordered nor greater than the next larger size shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

[2]The radial arm length of the drilling machine is defined as the horizontal distance between the column face and the centerline of the drill head spindle at its maximum travel distance from the column.

3.2.3.2 Quill and spindle assembly. Drill feed shall be accomplished by means of a hand-feed spoke lever and a rack and pinion gear arrangement. The spindle taper shall be a Morse taper and meet the requirements of ASME BS.10 and size requirements as shown in table II.

3.2.3.3 Power feed. The spindle shall have power feed requirements.

3.2.3.4 Depth stop. The machine shall have a depth stop equipped with a positive locking device. The depth stop shall be provided with an indicating scale marked in increments of not greater than 1/16 inch.

3.2.3.5 Drill chuck. A drill chuck shall be provided for the spindle with a 1/2-inch diameter drill capacity and taper shank that corresponds to the spindle taper of the size machine specified.

3.2.4 Controls. All controls shall be located for convenient access by the operator. The following controls, as a minimum, shall be provided:

a. On/off power switch.

b. Emergency stop button.

c. Coolant flow.

3.2.5 Lubrication system. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.6 Coolant system. A flood tool coolant system shall be provided. At minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The strainer shall be removable for cleaning. The coolant system shall have a means to control and direct the rate of clean coolant flow within the work area.

3.2.7 Electrical requirements. The electrical system shall conform to the National Fire Prevention Association (NFPA) 79. The machine shall have 220-volt, single-phase, 60-hertz circuit capability and shall be wired for the voltage specified. The control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.8 Drive motor. The spindle drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1. The motor shall have open drip-proof enclosure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

3.3 Performance. The drilling machine shall meet the alignment and performance requirements shown in tables III and IV, respectively, when tested in accordance with sections 5.3.1 through 5.3.3.

TABLE III. Alignment requirements.

Parameter	Maximum allowable deviation
Base and worktable flatness	0.010 inches
Spindle taper run-out TIR.[1] @ 1-1/4 inches from face TIR @ 5 inches from face	 0.002 inches 0.005 inches
Spindle (quill) travel Square with worktable Spindle incline toward column	 0.005 inches 0.005 inches
Worktable travel square to column	0.005 inches

1 TIR is the total indicated reading.

TABLE IV. Performance requirements.

Parameter	Tolerance
Vertical alignment	0.01 inch/foot, TIR[1]
Hole straightness	0.005 inch, TIR hole top to bottom
Hole maximum diameter	Test drill diameter +0.007 inch
Hole diameter out of roundness	0.005 inch, TIR

[1]TIR is the total indicated reading.

3.4 Safety and health requirements. The drilling machine shall be designed and manufactured in accordance with American National Standards Institute (ANSI) Bl 1.8. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.5 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be the US system of measurement.

3.6 Nameplate. A nameplate shall be securely attached to the machine. At minimum, the nameplate shall contain the following information:

a. Nomenclature.

b. Manufacturer's name.

c. Serial number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

d. Machine model designation.

e. Power Input (volts, total amperes, phase, frequency).

f. Short-circuit/over-current rating.

g. Contract number or order number.

h. National stock number.

i. Date of manufacture.

3.7 Lubrication plate or chart. A lubrication plate or chart shall be attached to the machine. The plate shall include the following information:

a. Points of lubricant application.

b. Servicing interval.

c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The drilling machine provided shall meet the salient characteristics of this commercial item description, conform to the manufacturer's drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The drilling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. The machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. The manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide a certification of calibration. The machine shall be tested in accordance with environmental conditions of ambient temperature between 40 and 104 °F and humidity between 20 and 95 percent, noncondensing.

5.3.2 Operational test. The drilling machine shall be operated in accordance with the manufacturer's standard operating procedures. The drilling machine shall be operated at no load for not less than 1/2 hour. Proper operation of all controls and features shall be verified during the trial period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

5.3.3 Performance tests. The performance tests shall be conducted on the drilling machine in accordance with section 5.3.3.2. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of part of the machine during or after the test.

5.3.3.1 Machine alignment. A machine alignment inspection shall be performed before and after the performance test using the criteria shown in table IV. Changes in machine alignment shall be cause for rejection.

5.3.3.2 Drill test. A test workpiece 1.0 inch in thickness and made of soft cast iron with a Brinell hardness of 120 to 150 shall be used. The test shall consist of drilling 5 holes through the test workpiece in 12 minutes. The drill bit nominal diameter shall be the maximum rated drilling diameter for the machine. Drill bit material, drill spindle speed, and feed rate shall be at the option of the manufacturer. The holes shall be measured and shall not exceed the values shown in table III for vertical alignment, hole straightness, maximum hole diameter, and out of roundness. The diameter of all holes shall be measured from the top to the bottom in three places. Out of roundness (the difference between the largest diameter of the hole and the smallest diameter of the hole) shall be determined by moving the measuring device 360° around the drilled hole.

5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables II and III. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING.

6. Preservation, packing, and marking. For acquisition purposes the floor mounted drilling machine shall be preserved, packed, and marked as specified in the acquisition order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Government documents. Electronic copies of specifications, standards, and handbooks are available at https://assist.dla.mil/.

7.1.2 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org/.

7.1.3 American Society of Mechanical Engineers (AMSE). Copies of ASME documents may be obtained at https://www.asme.org.

7.1.4 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.nema.org/.

7.1.5 National Fire Protection Association (NFPA). Copies of NFPA documents may be obtained at https://www.nfpa.org/.

7.1.6 National Institute of Standards and Technology (NIST). Copies of NIST documents may be obtained at https://www.nist.gov.

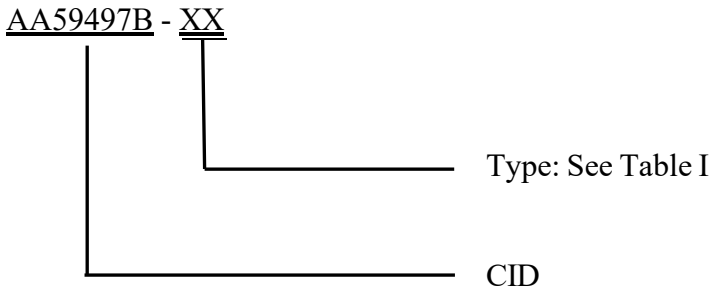
7.1.7 Occupational Safety and Health Administration (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.

7.1.8 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

7.1.9 Federal Acquisition Regulation (FAR). Copies of the FAR may be obtained at https://www.acquisition.gov/browse/index/far/.

7.2 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodian:
Army – AR
Navy – SH
Air Force – 184
DLA – GS

Preparing Activity:
Air Force – 184

Reviewer:
Army – SM
GSA – FAS

Agent:
Air Force – 110

(Project 3413-2025-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:15Z
Check the source to verify that this is the current version before use.